Technical Report on Design & Simulation of a Digital Stopwatch

Topic: Digital Stopwatch Implementation using IC 7490 & IC 7447

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1. Introduction

The Digital Stopwatch is a cornerstone project in digital electronics, serving as a practical application of sequential logic circuits. Unlike traditional mechanical or analog timepieces that rely on gear ratios and oscillating balances, a digital stopwatch utilizes electronic pulses generated by a stable oscillator to measure time intervals with extreme precision.

The core objective of this project is to design and simulate a stopwatch capable of counting from 00 to 59 seconds. This is achieved through the cascading of decade counters and the conversion of binary data into a human-readable format using BCD-to-7-segment decoders. This report details the architectural design, component selection, and logic transitions required to maintain a synchronized timing system.

2. Components Required & Descriptions

Component	Technical Description
IC 7490N	A versatile BCD decade counter consisting of four master-slave flip-flops. It is configured to provide a divide-by-two and a divide-by-five counter, which together produce a MOD-10 (0-9) counting sequence.
IC 7447N	An active-low BCD to 7-segment decoder/driver. It accepts 4-bit BCD input and provides the necessary logic to drive a common-anode 7-segment display by pulling the required segment pins to ground.
IC 7408N	A Quad 2-input AND gate IC. It serves as the primary control logic element, enabling the "Start/Stop" functionality by gating the clock signal and assisting in the reset logic for the tens digit.
Function Generator	Acts as the heart of the system by providing a stable 1 Hz square wave signal. This ensures that the stopwatch increments exactly once every second, maintaining the accuracy of the time measurement.
7-Segment Display	A common-anode LED display where all anodes are connected to VCC. Individual segments (a-g) are illuminated when the decoder provides a LOW signal to the corresponding cathode.
5V DC Power Supply	Provides the regulated voltage necessary for the Transistor-Transistor Logic (TTL) circuits to operate reliably within their defined logic levels.

3. BCD Truth Table (IC 7490 Decade Counter)

The table below represents the 4-bit Binary Coded Decimal (BCD) output sequence from the IC 7490 decade counter, showing the count progression from 0 to 9, followed by the reset to 0 for the new cycle.

Count	Q_A	Q_B	Q_C	Q_D
0-[Start]	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10 [New Cycle]	0	0	0	0

4. BCD to 7-Segment Truth Table

This table defines the logic states for segments **a** through **g** based on the BCD inputs A, B, C, and D (Common Anode configuration).

A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1

5. Working Principles

The operation of the digital stopwatch is based on the principle of frequency division and sequential counting. The system begins with a **Function Generator** providing a 1 Hz square wave. This signal is routed through an **AND Gate (IC 7408)**, which acts as a switch. When the control input is HIGH, the pulses reach the counter; when LOW, the counting pauses, effectively creating a "Start/Stop" mechanism.

The pulses are first received by the **Units Counter (IC 7490)**. This IC is wired in MOD-10 mode, meaning it cycles from 0 to 9. Upon reaching 9 and receiving the next pulse, it resets to 0 and generates a "Carry" pulse from its MSB output. This carry pulse serves as the clock input for the **Tens Counter (IC 7490)**.

The Tens Counter is monitored by logic gates to detect the number '6'. When the count reaches 60 (Units=0, Tens=6), the reset logic immediately clears both counters back to 00. Throughout this process, the **BCD-to-7-Segment Decoders (IC 7447)** constantly monitor the 4-bit binary outputs of the counters, translating them into the specific segment patterns required to display the digits visually on the 7-segment displays.

6. Applications

The logic implemented in this digital stopwatch is fundamental to numerous real-world systems. In **Academic Environments**, it serves as a primary tool for teaching students about clock synchronization, propagation delay, and IC interfacing. In **Sports and Athletics**, the same principle is used in high-precision race timers and scoreboards where elapsed time must be tracked and displayed instantaneously.

Furthermore, this circuit forms the basis for **Industrial Automation** timers, where processes must be triggered at specific intervals. It is also found in **Consumer Electronics**, such as microwave ovens, digital clocks, and parking meters. The modularity of the design allows for easy expansion—by adding more counter and decoder stages, the

system can be scaled to measure minutes, hours, or even milliseconds, making it a highly versatile architecture in digital system design.

7. Conclusion

In conclusion, the design and simulation of the Digital Stopwatch successfully demonstrate the efficiency of TTL-based sequential logic. By utilizing IC 7490 as a decade counter and IC 7447 as a decoder, we created a reliable system that accurately tracks time from 00 to 59 seconds. The integration of start/stop control via AND gates and the implementation of automatic reset logic highlight the precision that can be achieved through modular digital design. This project not only confirms the theoretical principles of BCD counting but also provides a robust foundation for developing more complex timing and synchronization systems in the future.

8. References

1. **Texas Instruments:** "SN74LS90 Decade Counter Datasheet," Technical Specifications and Pinout Diagrams.
2. **Texas Instruments:** "SN74LS47 BCD-to-Seven-Segment Decoder/Driver Datasheet," Logic Tables and Electrical Characteristics.
3. **Floyd, T. L.:** "Digital Fundamentals," 11th Edition, Pearson Education (Comprehensive guide on sequential logic).
4. **Electronics Tutorials:** "Binary Coded Decimal (BCD) and 7-Segment Display Interfacing," Online Resource.
5. **Multisim Documentation:** NI Multisim User Guide for Digital Circuit Simulation and Verification.